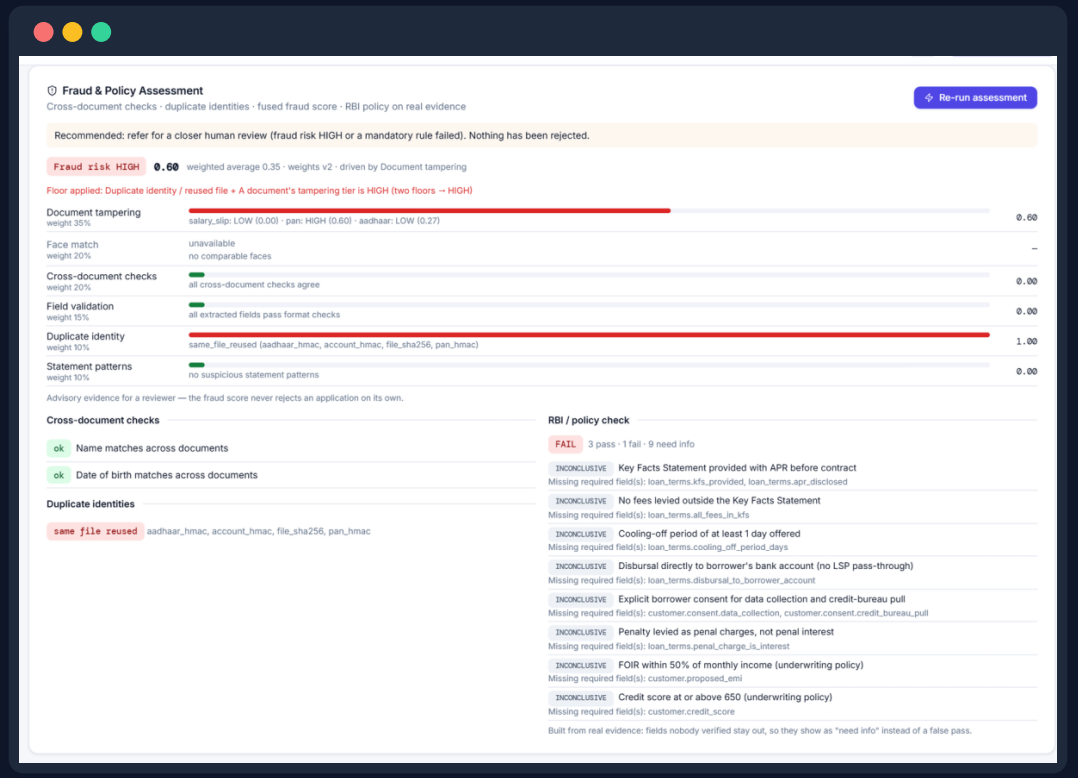
NBFC PLATFORM · PROJECT WALKTHROUGH

FinSight AI

Agentic loan underwriting for Indian NBFCs

29 CrewAI agents read the documents, check them for fraud and RBI policy, score the risk and draft the credit memo — on a local model, with a person making every decision.



29	6	12	61
agents	crews	loan products	automated tests

Shreyas Bile · 26 September 2026

CONTENTS

What this deck covers

Six parts, from what the product does to how well each agent works.

00

The product

The problem, the three users, a tour of the app

01

Architecture

Layers, tech stack, how one agent runs

02

The 29 agents

Six crews, what each agent does and how it scored

03

Workflows

Loan lifecycle, the end-to-end journey, a real run

04

Trust & control

Model gateway, guardrails, tracing, the database

05

Evaluation

Method, scorecard, defects fixed, latency, limits

00

The product

What FinSight does, who uses it, and what it looks like.



IN THIS SECTION

- The problem
- Three roles
- Twelve loan products
- App tour

THE PRODUCT

From a manual loan desk to agents that do the checking

Every NBFC loan file is a stack of Indian documents that someone has to read, verify and judge.

TODAY

Manual document reading

Officers retype PAN, Aadhaar, payslip and statement fields by hand.

Forgeries are hard to see

An edited payslip or a pasted ID photo looks genuine to the eye.

Slow, uneven turnaround

In the seeded process data a file waits 77 h on average before its documents are read.

Rules keep changing

RBI Digital Lending Directions 2025, KFS and penal-charge rules apply to every loan.

WITH FINSIGHT

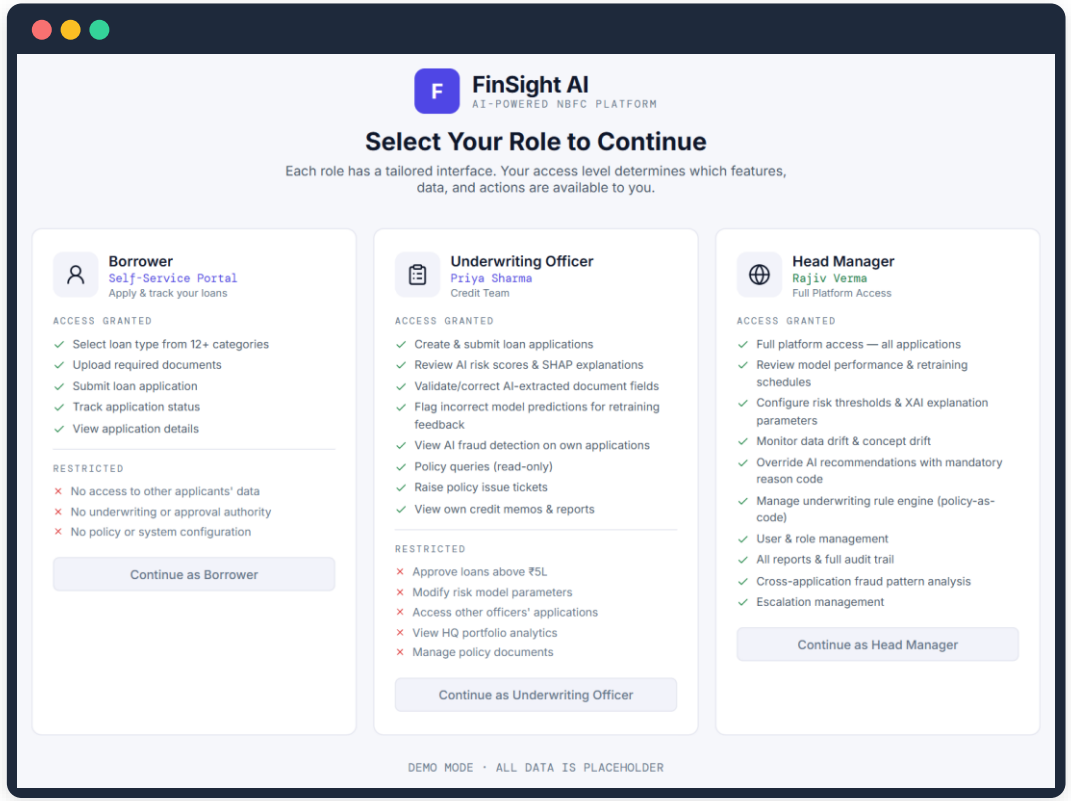
- Reads each document into fields and validates them
- Checks every document for tampering and fabrication
- Matches faces across documents, confirms a live selfie
- Fuses all fraud evidence into one advisory score
- Tests the case against RBI and internal rules
- Scores risk, drafts the credit memo, offer and next step

People decide. Nothing is auto-rejected, and any loan above ₹5 lakh needs the Head Manager's approval.

THE PRODUCT

Three roles, twelve loan products

One app, three views. Each product has its own document checklist and its own lifecycle branch.



Borrower

Applies, uploads documents, confirms what the agents read, takes a selfie and submits.

Underwriting Officer

Works the queue: evidence, fraud report, policy check, credit memo — then recommends.

Head Manager

Approves loans above ₹5 lakh, runs fraud triage, manages AI models and RBI rules.

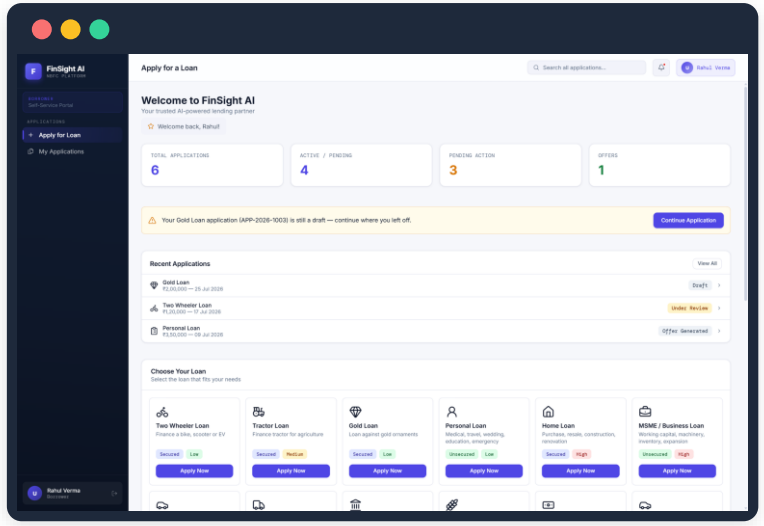
12 LOAN PRODUCTS

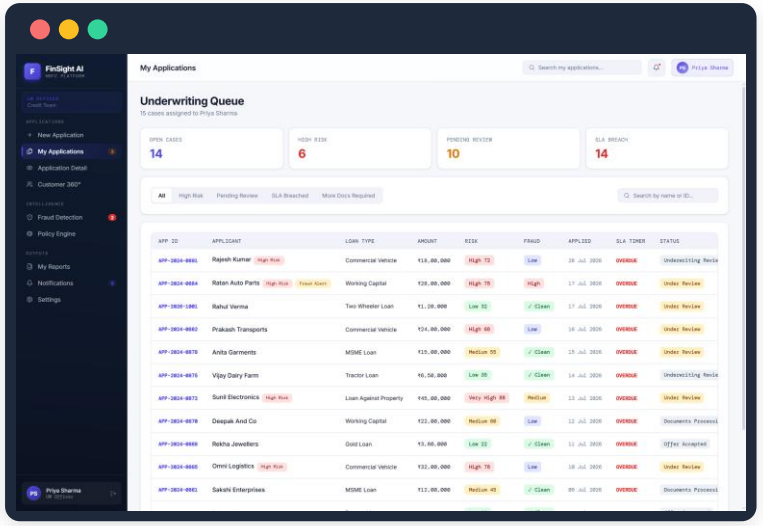
- Two Wheeler
- Tractor
- Gold
- Personal
- Home
- MSME / Business
- Car
- Commercial Vehicle
- Loan Against Property
- Microfinance / Rural
- Working Capital
- Used Vehicle

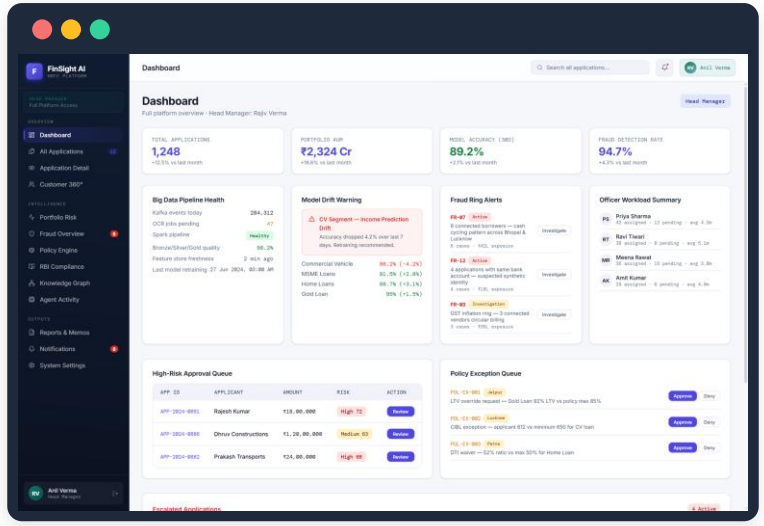
THE PRODUCT

A quick tour of the app

Screens captured from the running app (the application data shown is seeded demo data).







Borrower portal

Status of every application and the 12 products to apply for.

- Apply for any of the 12 products
- Upload, confirm fields, take a selfie
- Track status and offers

Officer queue

Assigned cases with risk, fraud, SLA timer and next action.

- Risk and fraud column for every case
- SLA timer and the next action
- Open a case: 8 tabs of evidence

Head dashboard

Portfolio view: approvals waiting, fraud alerts, model health.

- Approvals waiting above ₹5 lakh
- Fraud alerts and ring triage
- AI model and RBI rule control

01

Architecture

Six layers, one gateway to the language model, and one rule for every agent: tools decide, the model only explains.

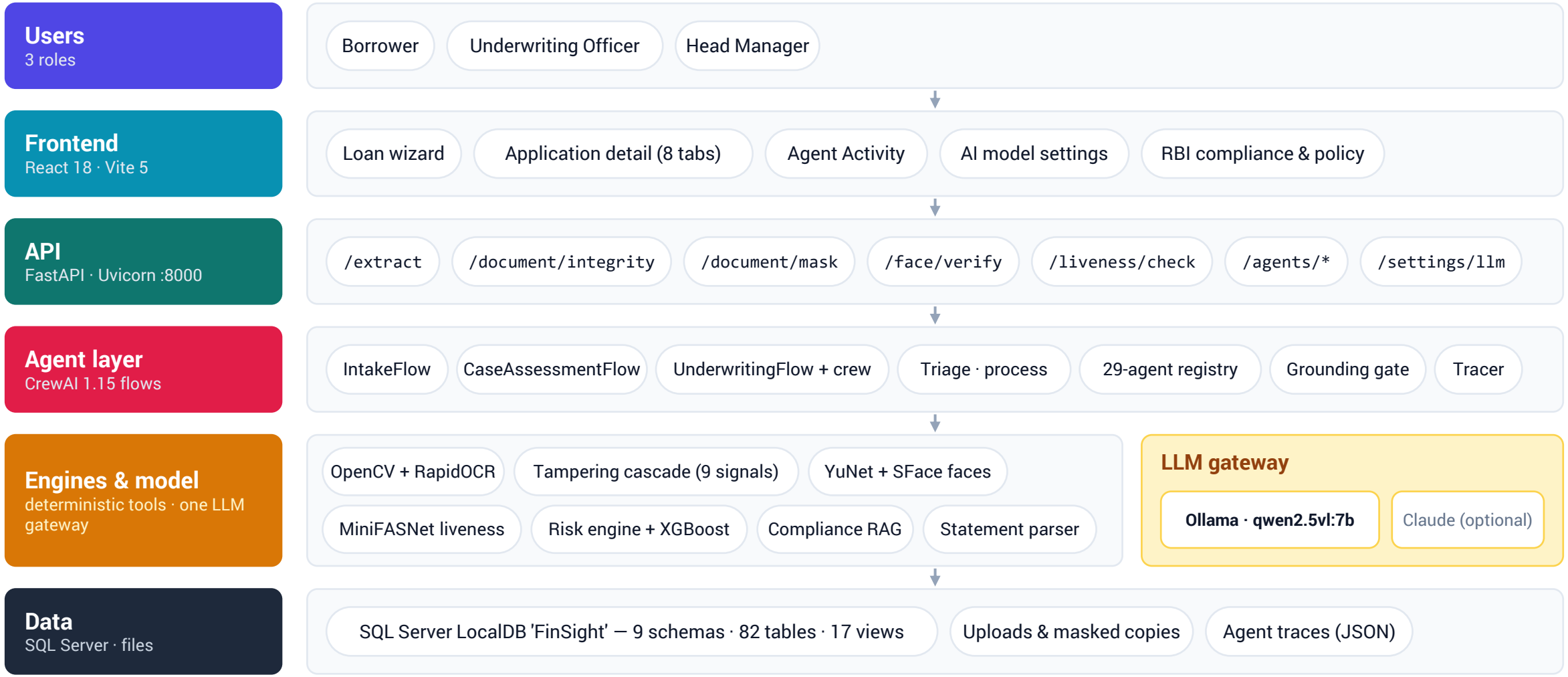


IN THIS SECTION

System layers | Tech stack | How one agent runs

ARCHITECTURE

System architecture



ARCHITECTURE

Tech stack

Everything runs on one laptop — the local model uses an RTX 4060 (8 GB).

Frontend
React 18 · Vite 5 · lucide icons
Role-based single-page app

API
FastAPI 0.115 · Uvicorn · Pydantic 2
Python 3.13

Agents
CrewAI 1.15.22
Flows, crews and 29 registered agents

Language models
Ollama · qwen2.5vl:7b (default)
qwen3-vl:4b option · Claude optional

Vision & OCR
OpenCV 4.10 · RapidOCR (ONNX)
PDFium (pypdfium2) for PDFs

Face & liveness
YuNet · SFace · MiniFASNet
ONNX Runtime 1.27, all local

ML & scoring
XGBoost 3.3 · scikit-learn
Risk overlay trained on 50,000 NBFC loans

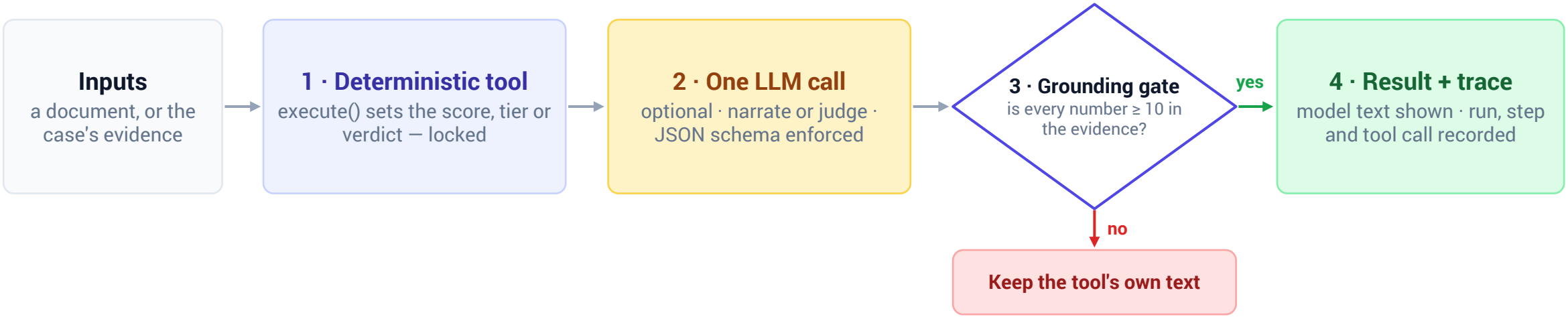
Data
SQL Server LocalDB · pyodbc
9 schemas · 82 tables · 17 views

Quality
pytest · evaluation harness
61 tests · 29 agent evaluations

ARCHITECTURE

How one agent runs: tools decide, the model only explains

Every agent follows the same four steps. A score is fixed before any language model is involved.



THREE KINDS OF AGENT

18 **Deterministic**

Tool only — no model at all.

Validation, masking, face match, tampering signals, fraud fusion, policy rules, risk, offer…

7 **Hybrid**

Tool decides, model narrates or judges within limits.

Authenticity, tamper explainer, fraud patterns, triage, rule authoring, income, process mining

4 **LLM**

The model does the core task inside a fenced call.

Extraction, regulatory Q&A, credit memo, underwriting supervisor

02

The 29 agents

Six crews. Each agent has one job, one set of tools and a pass mark fixed before it was tested.



IN THIS SECTION

Agent map | Intake | Privacy & identity | Tampering | Fraud fusion | Patterns & triage | Compliance | Underwriting | Risk | Operations

THE 29 AGENTS

The agent map: 29 agents in six crews

Intake
4 agents

- #1 deterministic — Preprocessing
- #2 LLM — Extraction
- #3 deterministic — Validation
- #4 deterministic — Document Identity

Privacy
1 agent

- #5 deterministic — PII Masking

Identity & fraud evidence
11 agents

- #6 deterministic — Face Verification
- #7 deterministic — Tampering Forensics
- #8 hybrid — Authenticity
- #9 hybrid — Tamper Explainer
- #10 deterministic — Tamper Locator
- #11 deterministic — Cross-Doc Verification
- #12 deterministic — Dedupe
- #13 deterministic — Fraud Fusion
- #14 hybrid — Fraud Patterns
- #15 hybrid — Fraud Triage
- #29 deterministic — Liveness

Compliance
4 agents

- #16 deterministic — KB Ingestion
- #17 hybrid — Rule Authoring
- #18 LLM — Regulatory Q&A
- #19 deterministic — Policy Check

Underwriting
6 agents

- #20 hybrid — Income Analysis
- #21 deterministic — Risk Scoring
- #22 LLM — Credit Memo
- #23 LLM — Underwriting Supervisor
- #24 deterministic — Offer Generation
- #25 deterministic — Workflow Router

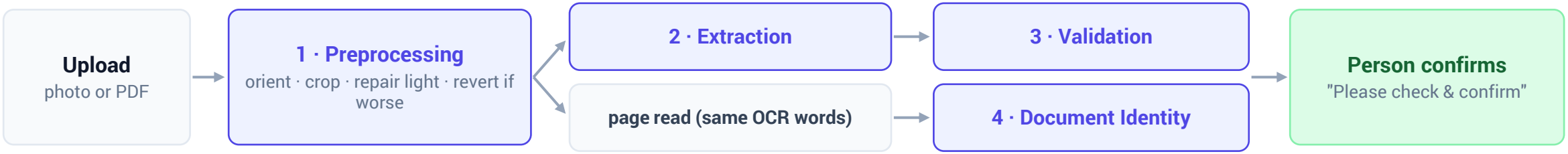
Operations
3 agents

- #26 deterministic — Notifier
- #27 deterministic — Audit / Storage
- #28 hybrid — Process-Mining Analyst

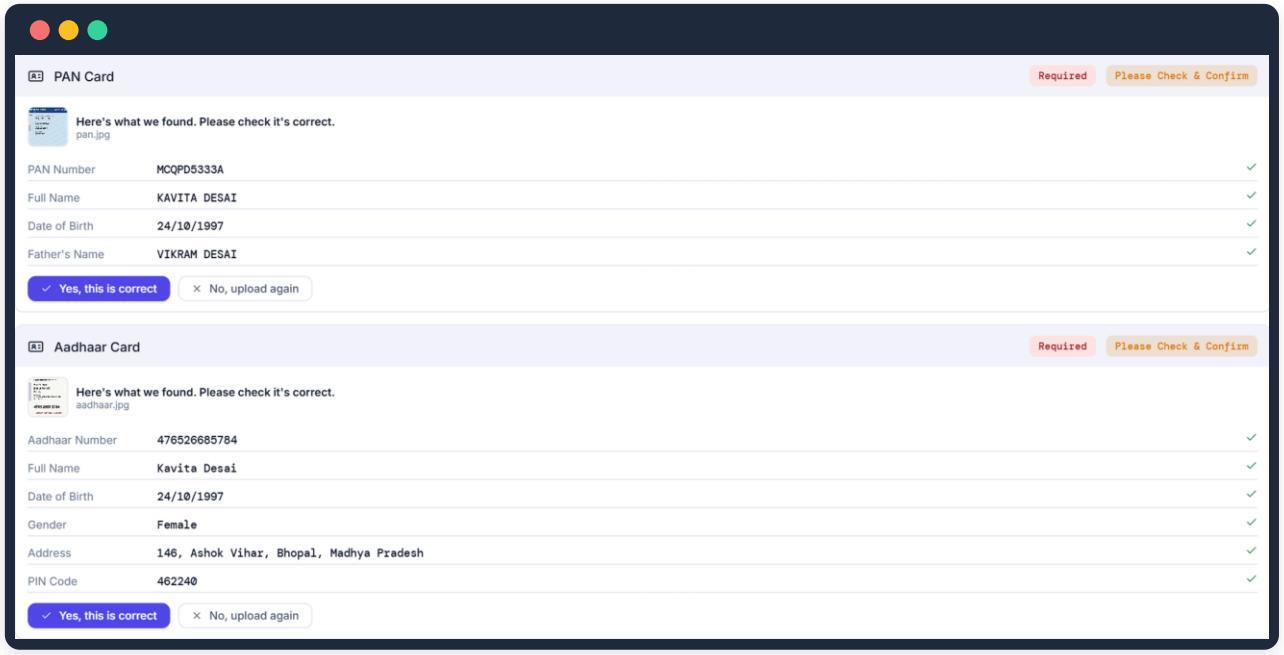
THE 29 AGENTS · INTAKE CREW

Intake: every upload becomes checked, structured fields

Four agents run on each document, in about 10 seconds on the local model.



qwen2.5vl reads every requested field into JSON, inventing nothing · Verhoeff, PAN, IFSC, GSTIN formats · the page's own phrases must match the requested type



99.6% — **Preprocessing**
field text readable on 60 degraded cards

97.2% — **Extraction**
field accuracy · 48 docs, 318 fields · 9.1 s

100% — **Validation**
1,487 generated IDs flagged correctly

100% — **Document Identity**
wrong uploads caught (was 54%)

Privacy & identity: mask the numbers, match the faces

PII Masking protects the officer's view; Face Verification and Liveness make sure one real person is applying.

PII Masking #5

PAN

~~MCQPD5333A~~ → XXXXXX333A

AADHAAR

XXXX XXXX 5784

- OCR finds the PAN / Aadhaar number and masks all but the last four
- Reads the card upright first, so sideways photos can't leak
- Officers and the audit view only ever see the masked copy

0 leaks — 48 test cards + 4 real scans · last four kept 93.8%

Face Verification #6

YuNet → SFace → ≥ 0.363

- YuNet finds each face, SFace turns it into an embedding
- Cosine similarity ≥ 0.363 → same person
- A blurry or tiny face can only be 'uncertain', never 'mismatch'

99.4% — correct verdicts on 160 LFW pairs · 0.3 s

Liveness #29

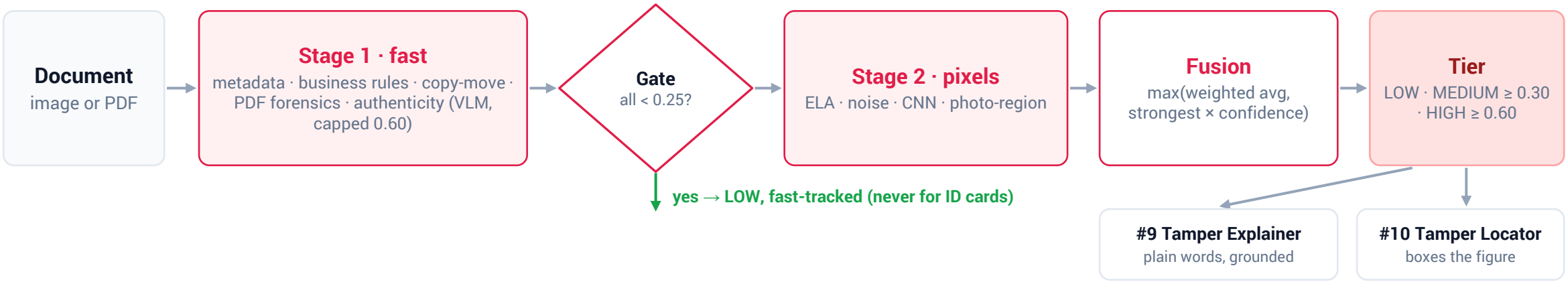
Selfie → MiniFASNet → ID match

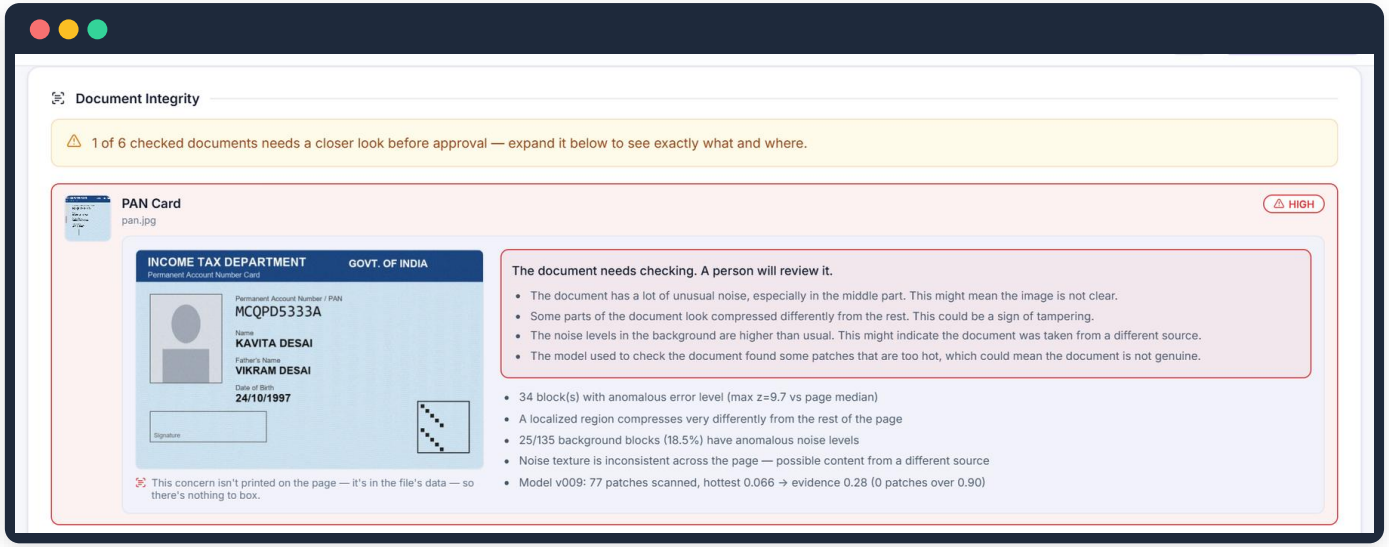
- Selfie checked by MiniFASNet for print / screen attacks
- Then matched to the ID photo on the case
- The picture is checked and discarded

22.2% — ACER (target ≤ 10%) — Weak; needs real selfies

Tampering: a nine-signal cascade, fused into one honest tier

Cheap checks first; pixel forensics only when needed (ID cards always get them). Two helper agents explain and locate.





100% — **Forgeries flagged**
72 forgeries, 7 attack types

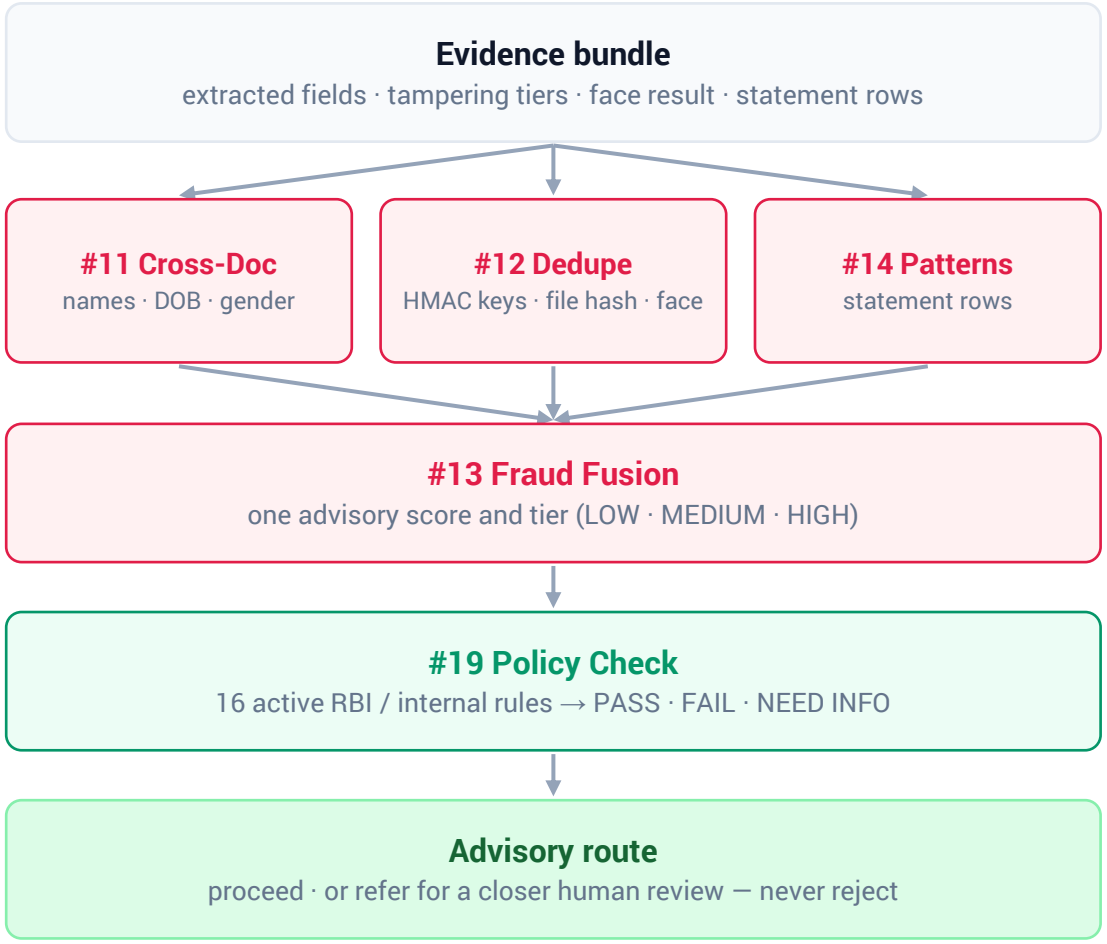
16.7% — **Genuine photos HIGH**
known over-flag on camera photos

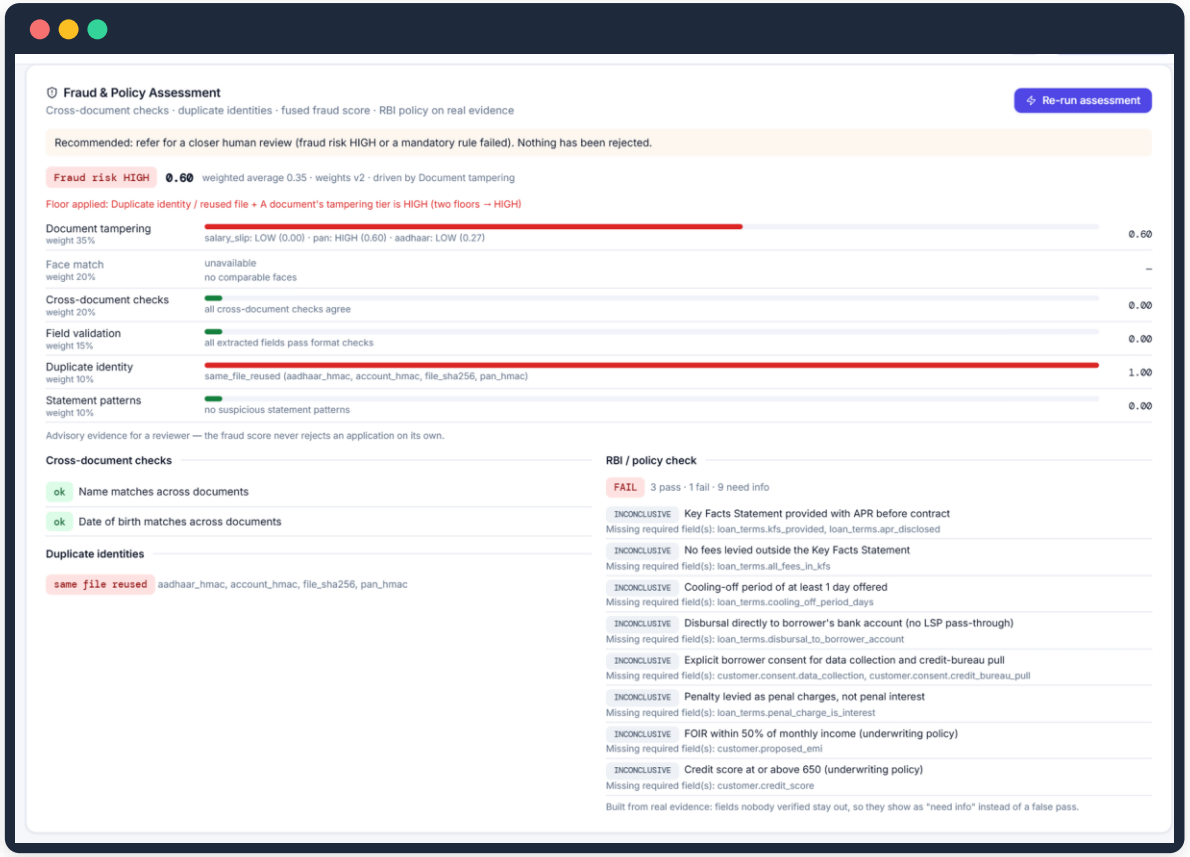
89.5% — **Authenticity · Explainer**
fabrications caught · explainer 90% grounded

Tamper Locator boxes only edited figures: 33% (Weak).

Case assessment: all the evidence, one advisory route

Runs when the file is complete — no language model at all, about 100 ms.



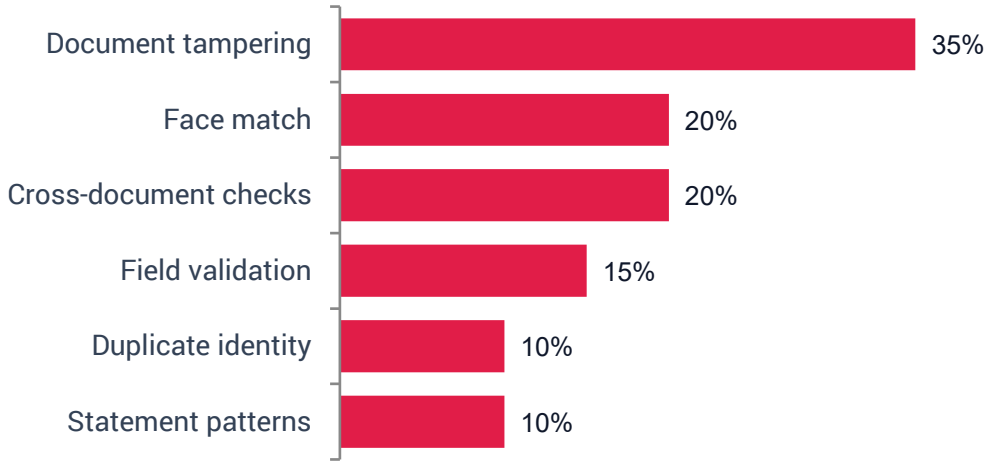


Real run, synthetic applicant submitted twice: Dedupe finds the same files → two floors → HIGH.

Fraud Fusion v2: how one score is built

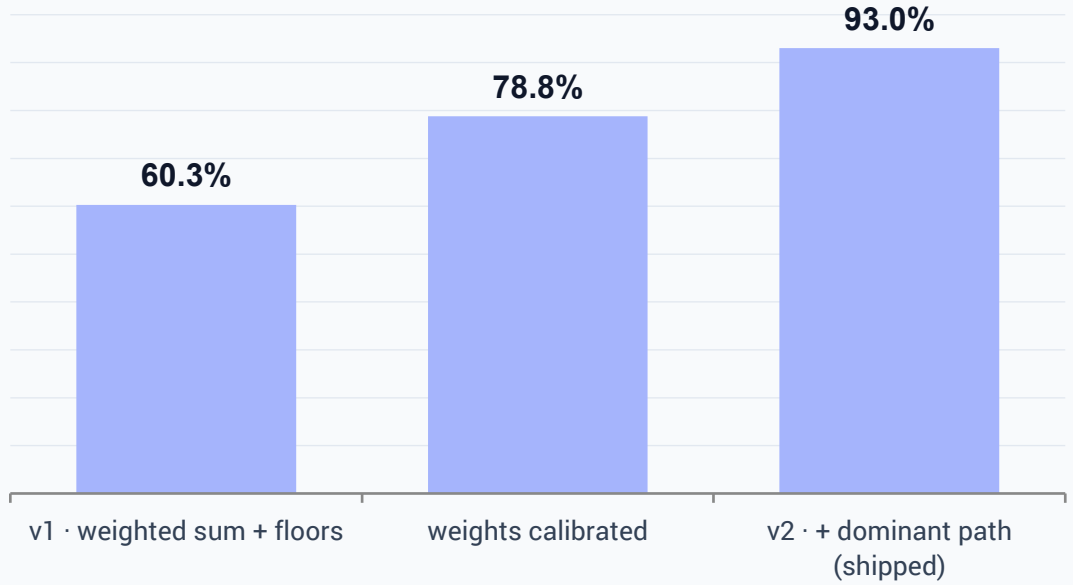
A weighted average measures agreement; a dominant path and floors stop one strong red flag being averaged away.

Component weights (normalised over what's available)



1. **Weighted average** Each component scores 0–1; only components with evidence count.
2. **Dominant path** fused = max(average, strongest signal × its confidence) — plus 0.1 if another signal agrees.
3. **Floors** Confident face mismatch · exact duplicate identity · tampering HIGH. One floor → at least MEDIUM; two → HIGH.
4. **Tiers, advisory only** MEDIUM ≥ 0.20 · HIGH ≥ 0.60. The score never rejects on its own.

Held-out tier accuracy · 400 labelled scenarios



93% meets the ≥ 90% target. In v2's errors, 17 of 35 HIGH cases were called MEDIUM — it leans cautious toward review, not toward rejection.

Statement patterns and fraud triage

Patterns read the bank statement row by row; triage looks across all applications for rings.

#14 FRAUD PATTERNS

PDF text layer
or OCR words

→

Tilt-corrected rows
date + amounts

→

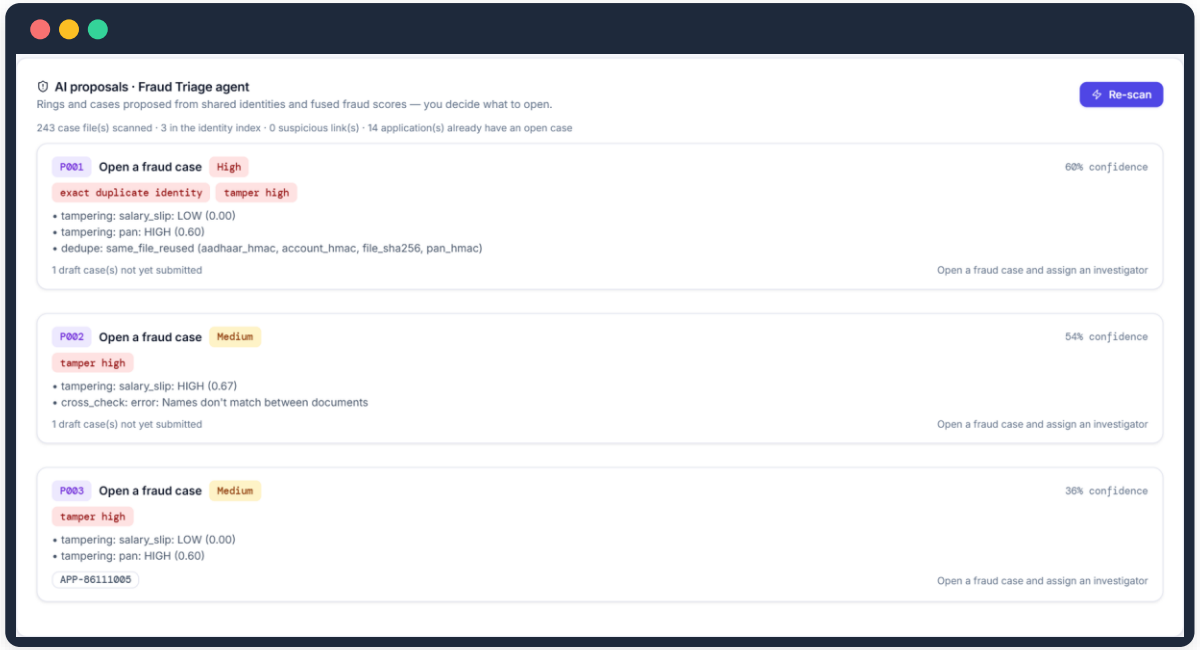
Reconcile
debit / credit by balance

FIVE PATTERNS

- **Cash cycling** money in and straight back out
- **Sudden deposit** a lump sum just before applying
- **Window dressing** balance pumped near statement end
- **Income mismatch** salary credits ≠ payslip net
- **GST inflation** credits far below declared turnover

F1 1.00

Pattern detection · 54 statements
photo statement rows parsed 99.5% (was 66%)

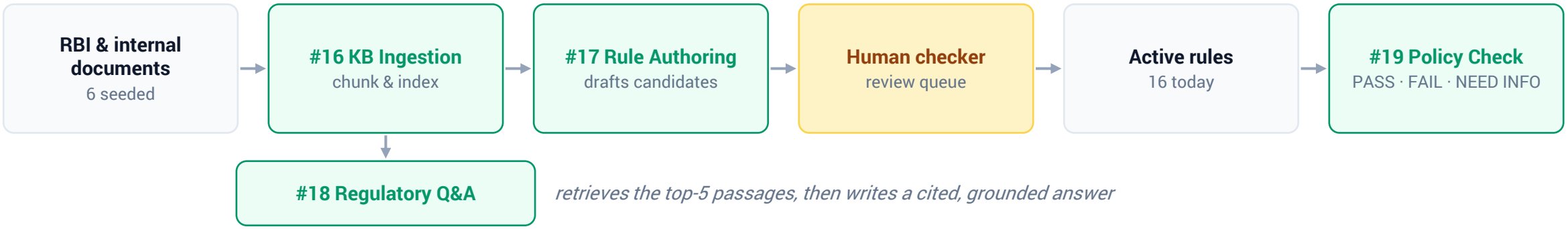


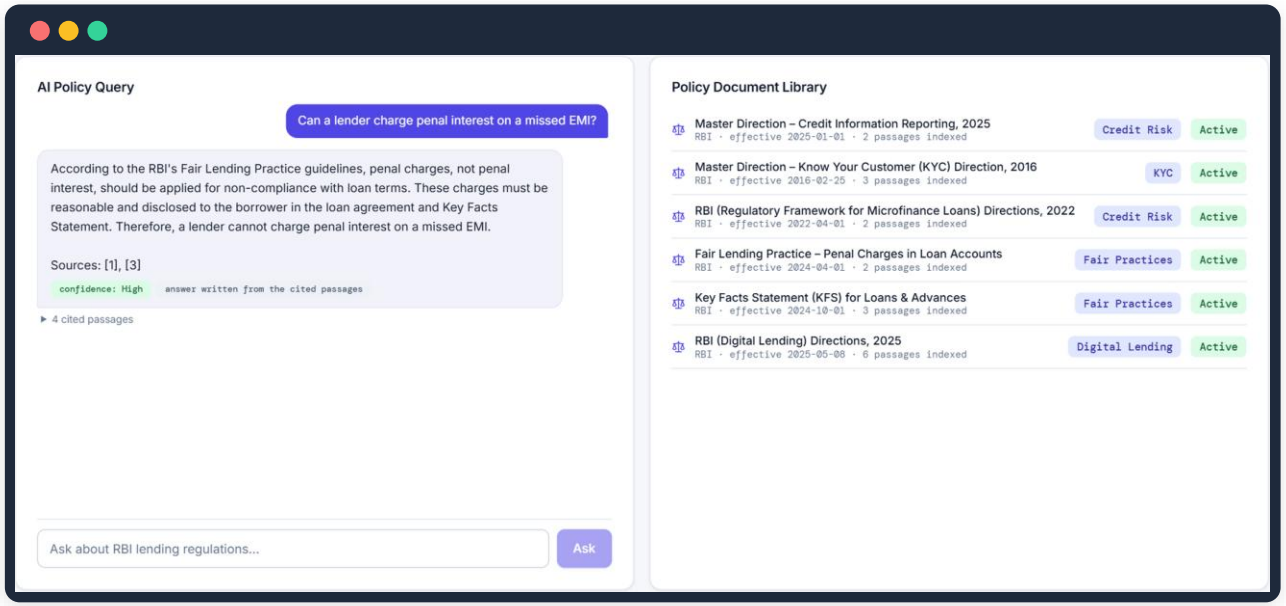
#15 FRAUD TRIAGE

- Links applications that share one face under several PANs, one bank account, one PAN with two faces, or one file
- Proposes cases only — a person decides what to open
- 9 of 9 planted rings and cases found, 0 false ones; repeat customers never flagged

Compliance: RBI rules as data, answers with citations

Regulations are indexed once; rules are drafted by an agent and approved by a person; every case is checked against them.





91.7% — **KB Ingestion**
recall@5 on 12 gold questions

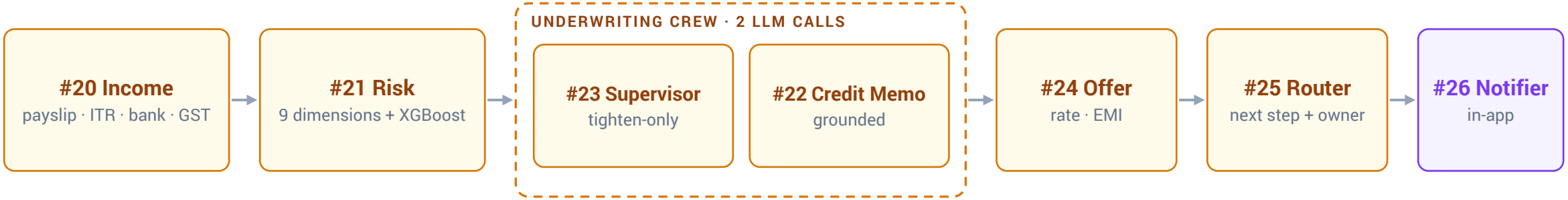
8 of 9 — **Rule Authoring**
known rules recovered (88.9%)

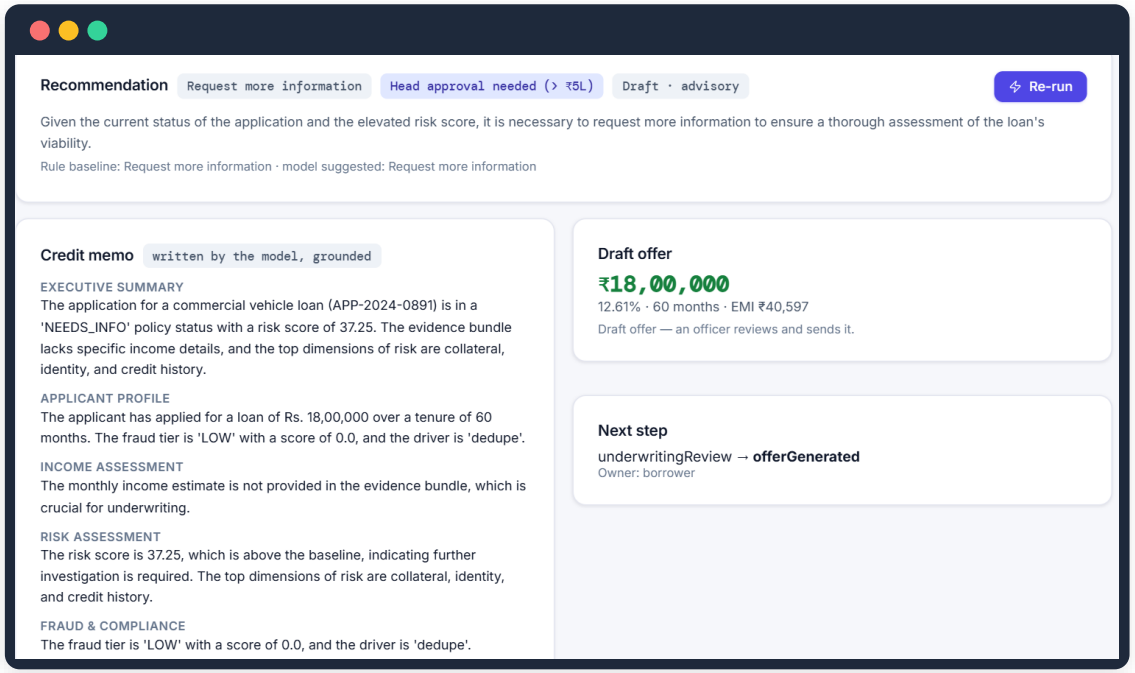
100% — **Regulatory Q&A**
answering passage in top 5 · 5.1 s

100% — **Policy Check**
violations detected one at a time

Underwriting: from income to a drafted offer

One click runs the chain; the only model calls are the two agents inside the crew. About 20 seconds.





F1 1.00 — **Income Analysis**
mismatch detection on 120 cases

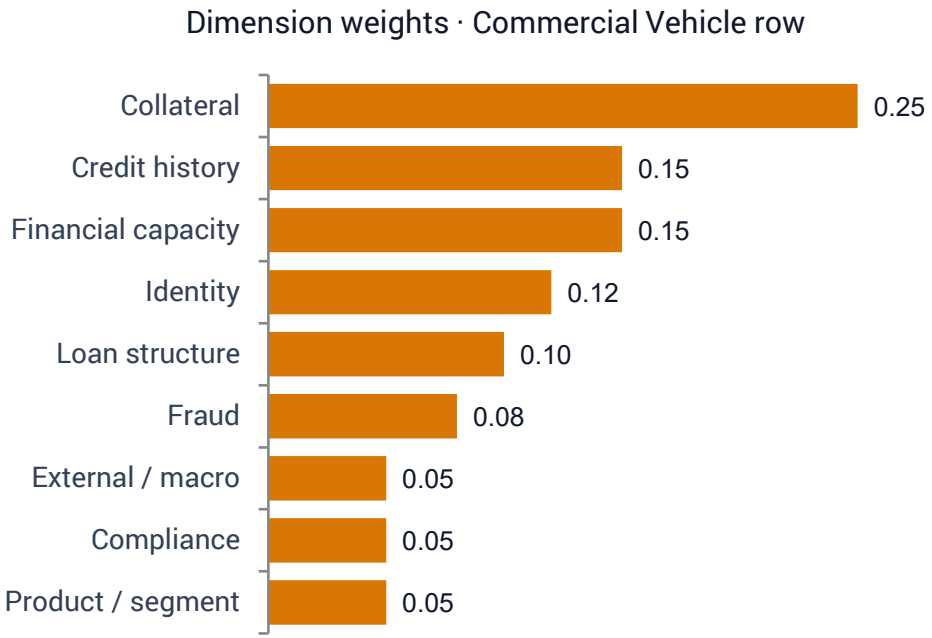
100% — **Credit Memo**
memos citing only real figures · 19 s crew

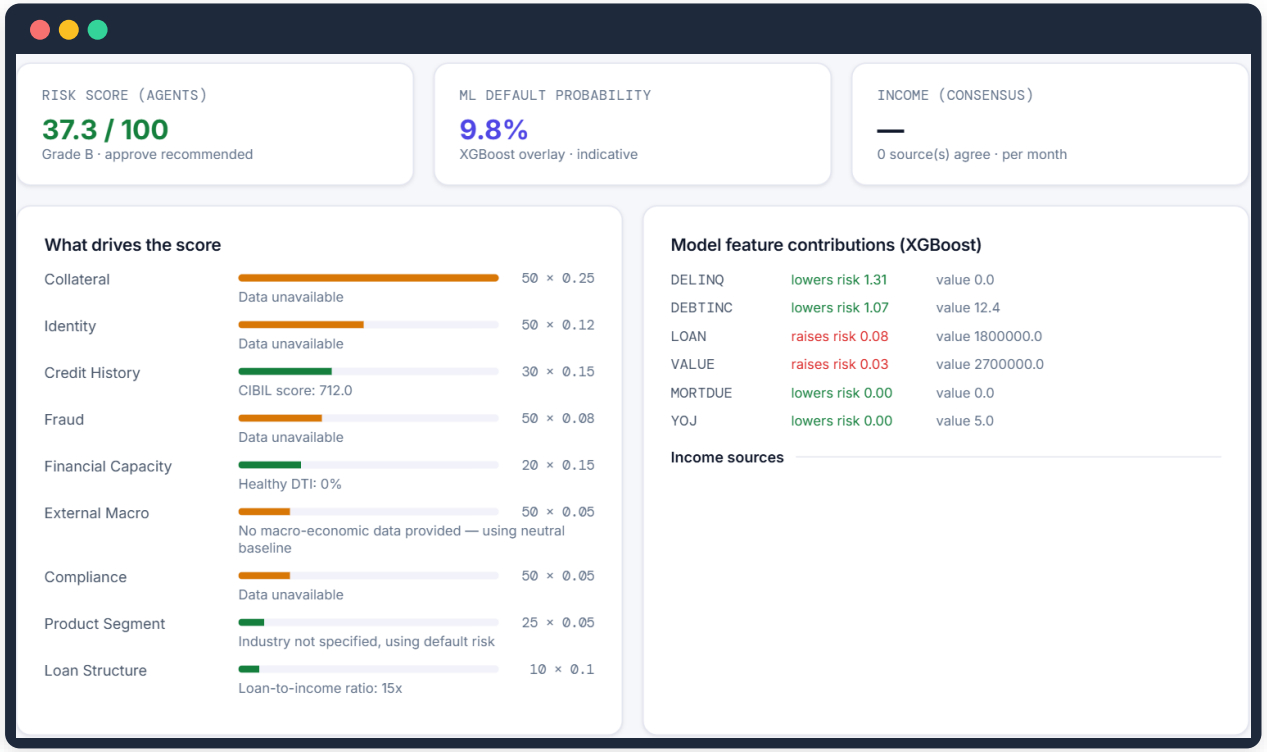
100% — **Supervisor**
agrees with policy · 1 loosening blocked

100% — **Offer · Router**
EMI = formula (864) · next step (111)

Risk Scoring: nine dimensions plus a machine-learning check

A 12 × 9 weight matrix (one row per product) gives the grade; XGBoost adds an indicative default probability.





Application › Risk Analysis: agent score, ML default probability and what drives each.

0.669 → 0.771
held-out ROC-AUC after the feature fix (target ≥ 0.75)

0 of 750
checks where a worse input lowered the risk score

Operations: notify, keep the record, watch the process

Three back-office agents. None of them can send anything outside the app.

#26 Notifier

Drafts notification rows from templates for borrower, officer and head.

100% correct recipients · 0 outbound attempts

#27 Audit / Storage

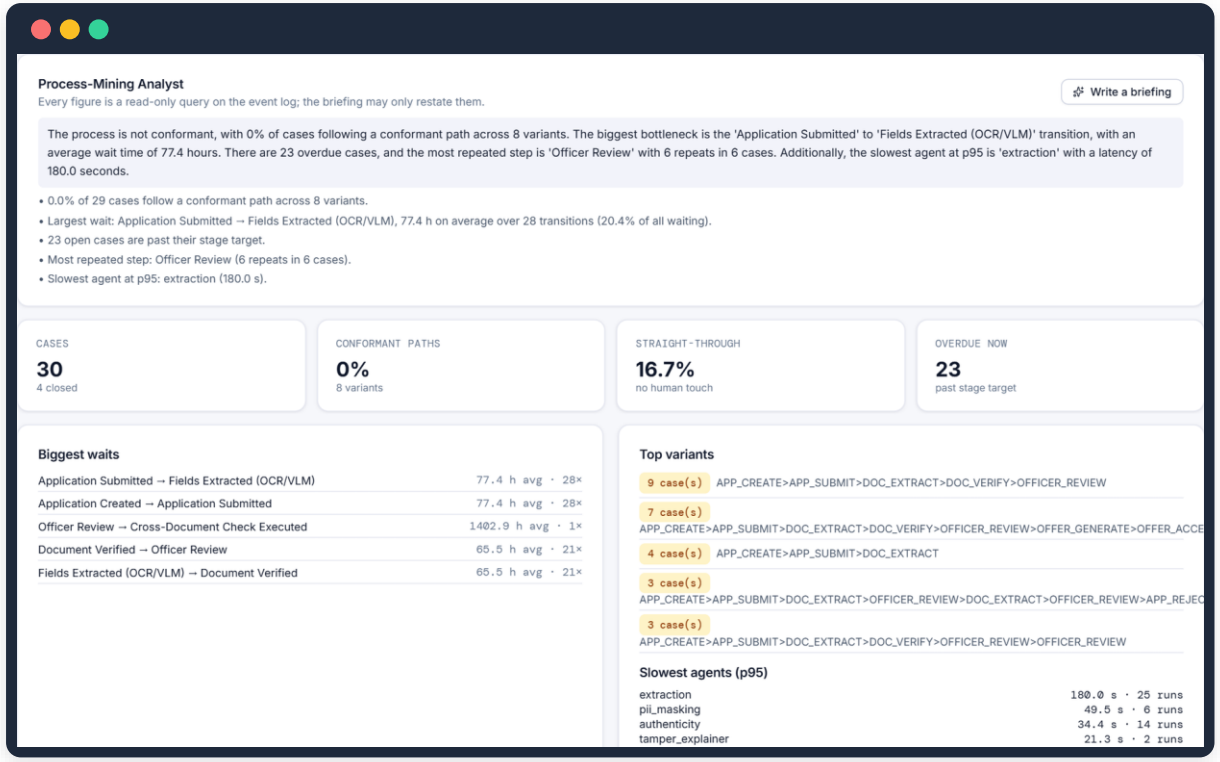
Persists files and SQL rows; officers see a masked copy.

100% files and rows intact

#28 Process-Mining Analyst

Conformance, variants, waits and slow agents from pm.* views; optional grounded briefing.

100% of figures equal direct SQL · 129 ms



Agent Activity › Process Mining — seeded demo process data: 0% conformant paths, 23 cases overdue.

03

Workflows

How a loan moves through its lifecycle, who acts at each step, and what a real run looks like.



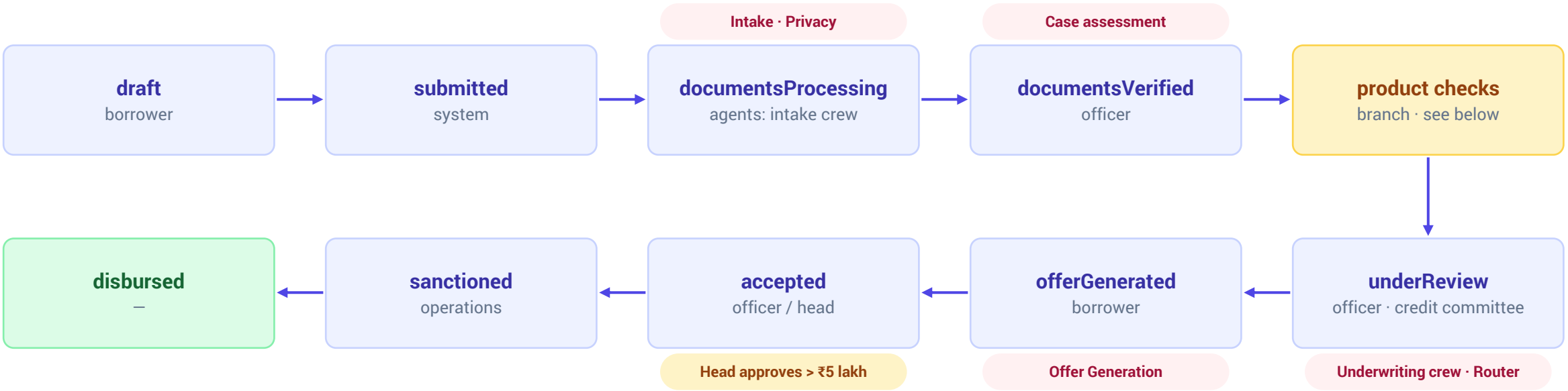
IN THIS SECTION

Loan lifecycle | End-to-end journey | A real run

WORKFLOWS

The loan lifecycle

One path for every product plus its own branch. The Workflow Router only recommends; a person moves the status.



PRODUCT BRANCHES (between documentsVerified and review)

Gold valuation scheduled → valued

Tractor · Microfinance field verification

Loan Against Property property valuation → legal

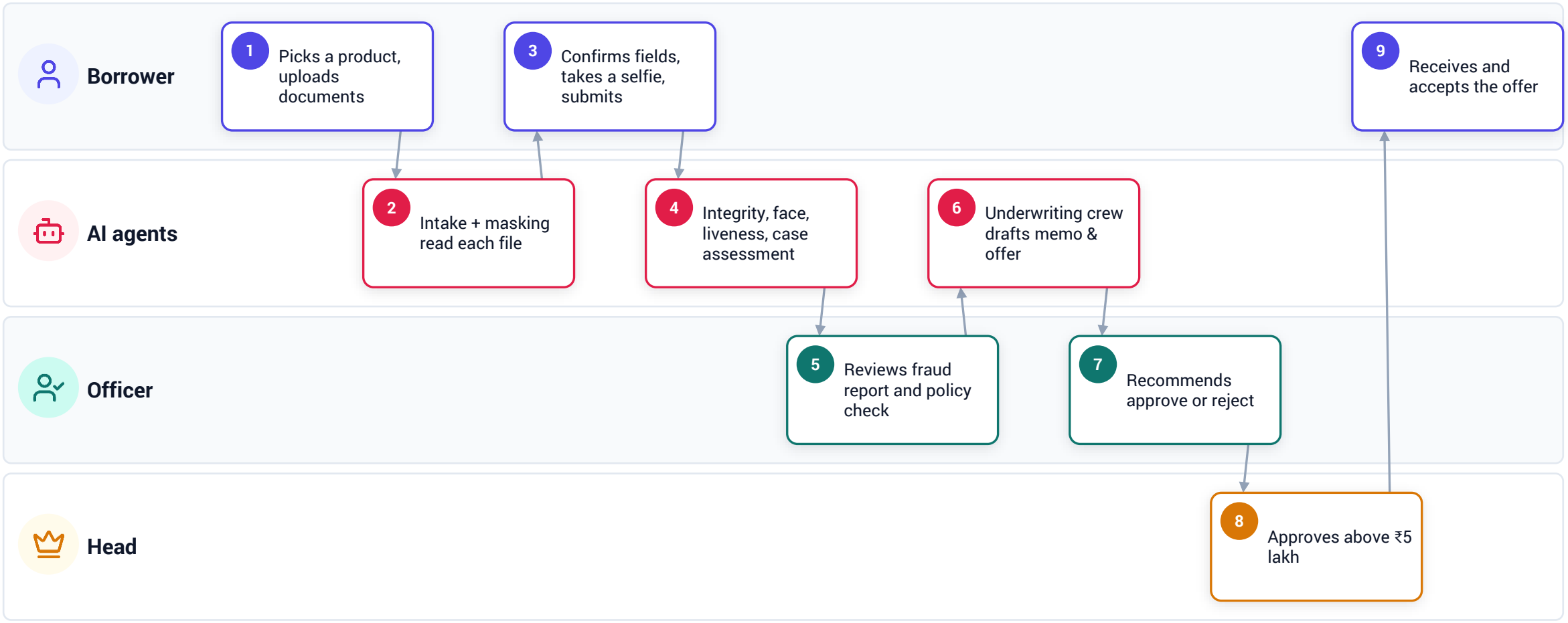
Home 6 checks: property, legal, income, docs, technical

MSME · Working capital bank & GST analysis → committee

Used vehicle vehicle inspection

End to end: who does what

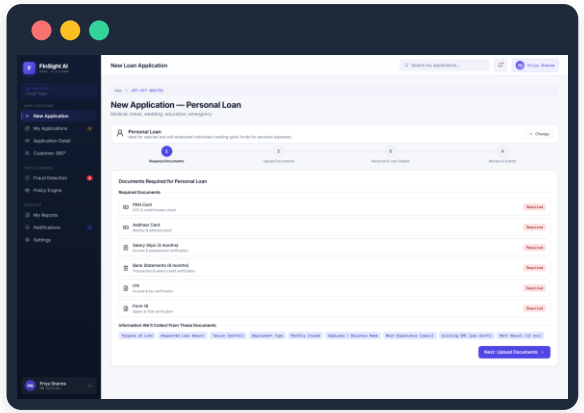
Agents work between the human steps; every agent output is evidence for the next person.



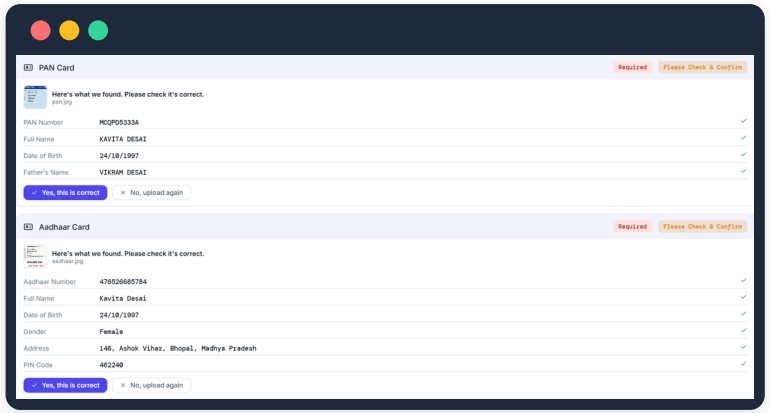
A real run through the wizard

Captured from the app with a synthetic applicant (generated identity — no real personal data).

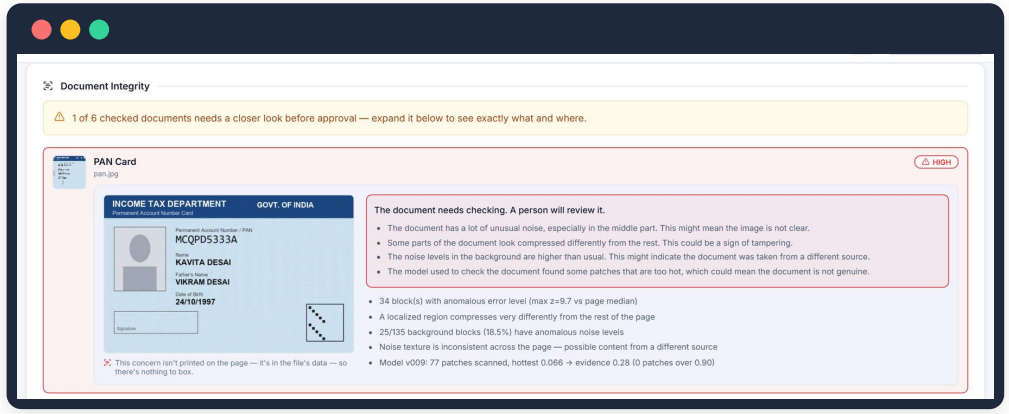
1 Product checklist



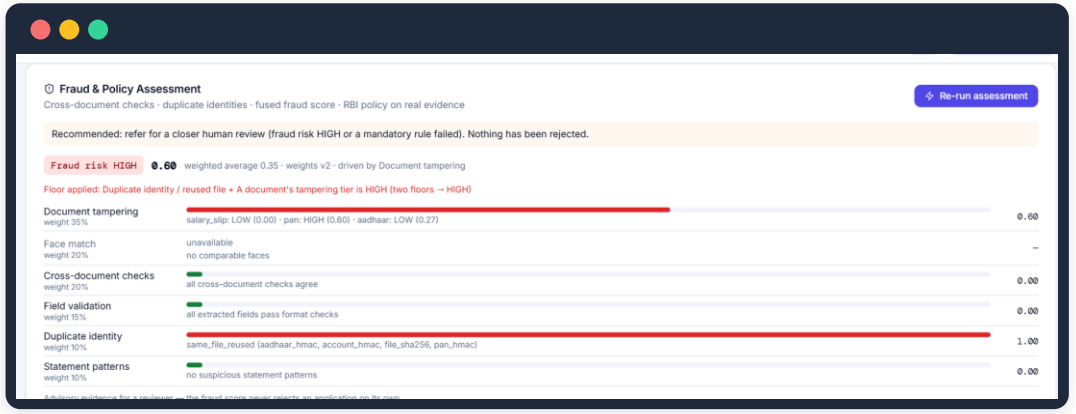
2 Fields read by the intake crew



3 Integrity: plain-language findings



4 Fraud & policy assessment



04

Trust & control

One gateway to every model, guardrails that are tested, and a trace of every step in SQL Server.



IN THIS SECTION

LLM gateway | Guardrails | Tracing | Data model

TRUST & CONTROL

One gateway for every model call

The Head Manager picks the model per agent, per lane or by default. The gateway enforces privacy and queues the calls.

Agent asks for a model
gateway.chat(agent_id, …)

↓

1 · Resolve the model
agent override → vision / text lane → default · snapshot per run

↓

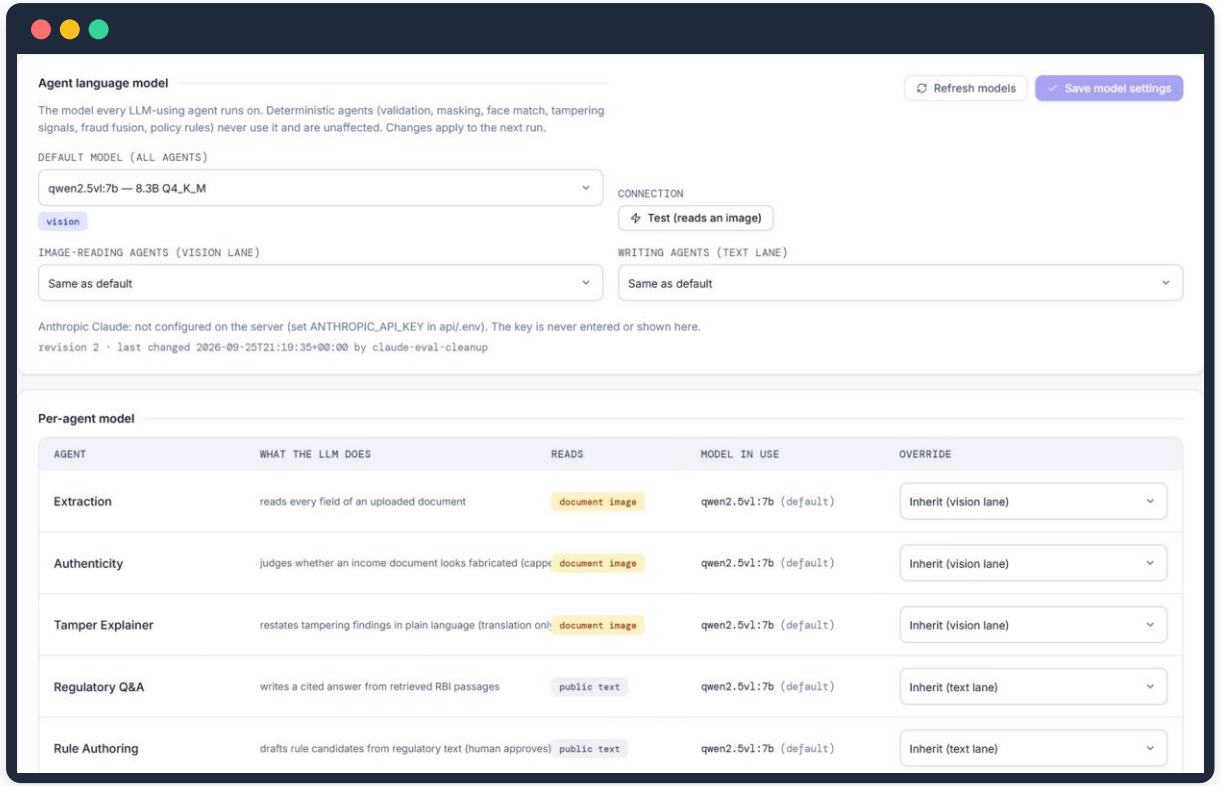
2 · Privacy guard
FORBID_HOSTED switch · Head acknowledgement · pseudonymised text for hosted

↓

3 · LLM gate
one slot per provider · priority queue · 180 s max wait

Ollama (local)
127.0.0.1 · pooled · images fitted to context

Anthropic Claude
optional · not configured today



System Settings › AI Models (Head only): default, lanes and a per-agent override table.

TRUST & CONTROL

Guardrails — each one pinned by a test

The promises the design makes, and how each was verified.

LLM never sets a score

Scores, tiers and verdicts come from tools only.

tier unchanged 40 / 40

Grounding gate

Every number ≥ 10 in model text must be in the evidence.

ungrounded text dropped

Tighten-only supervisor

May make the next step stricter, never looser.

1 loosening attempt blocked

Nothing auto-rejects

Advisory routes only; above ₹5 lakh needs the Head.

router guard test

No side-effect tools

No agent can approve, reject, file, accept or send.

checked on all 29

LLM budget per path

/extract ≤ 1 call; masking, face and assessment 0.

call-count test

Authenticity capped

The fabrication signal is capped at 0.60, never alone.

0 lone decisions / 52

Hosted models gated

Head acknowledgement, kill switch, pseudonymised text.

settings + gateway tests

Identity keys hashed

Dedupe stores HMAC digests, never raw PAN / Aadhaar.

holds

Contracts unchanged

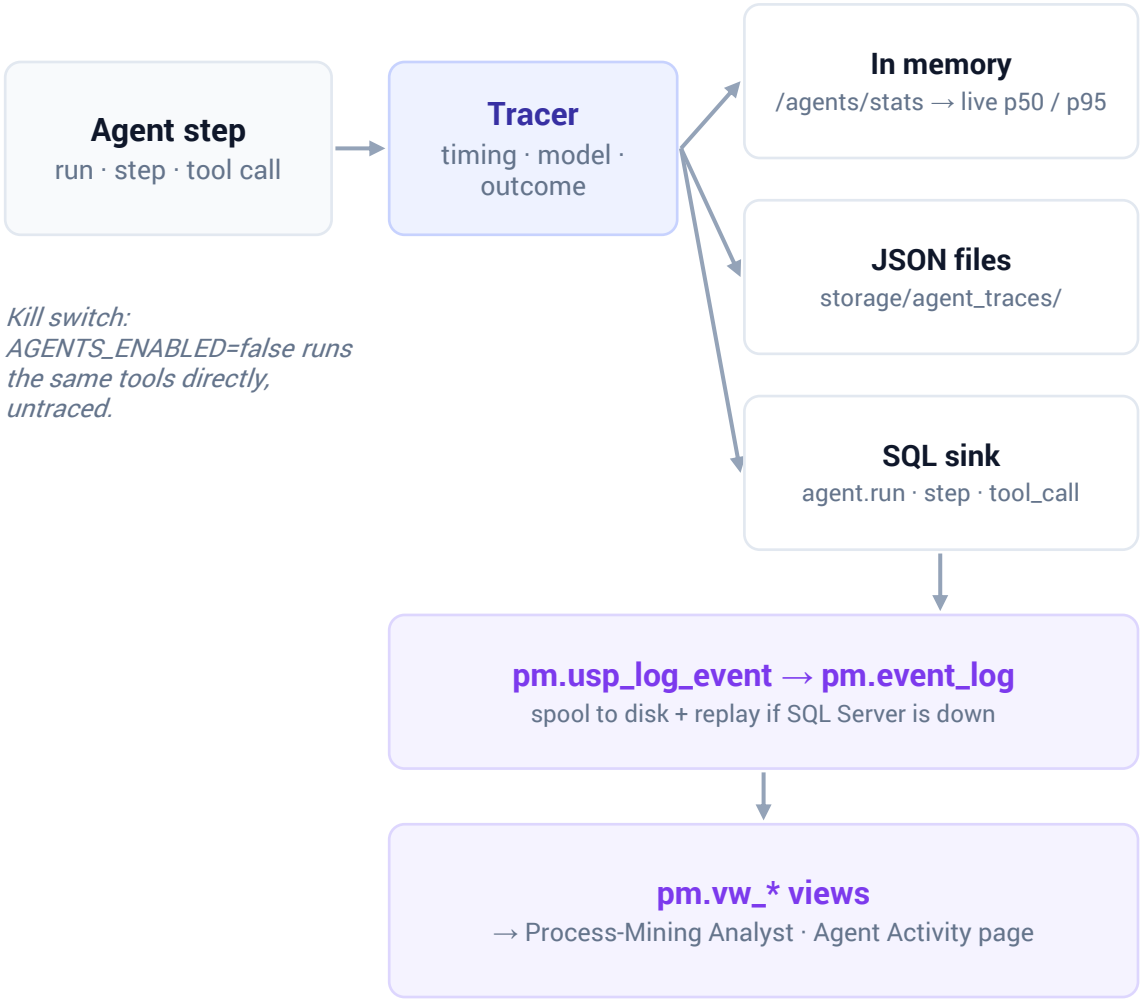
Pre-agent endpoints answer exactly as before.

24 / 24 contract tests

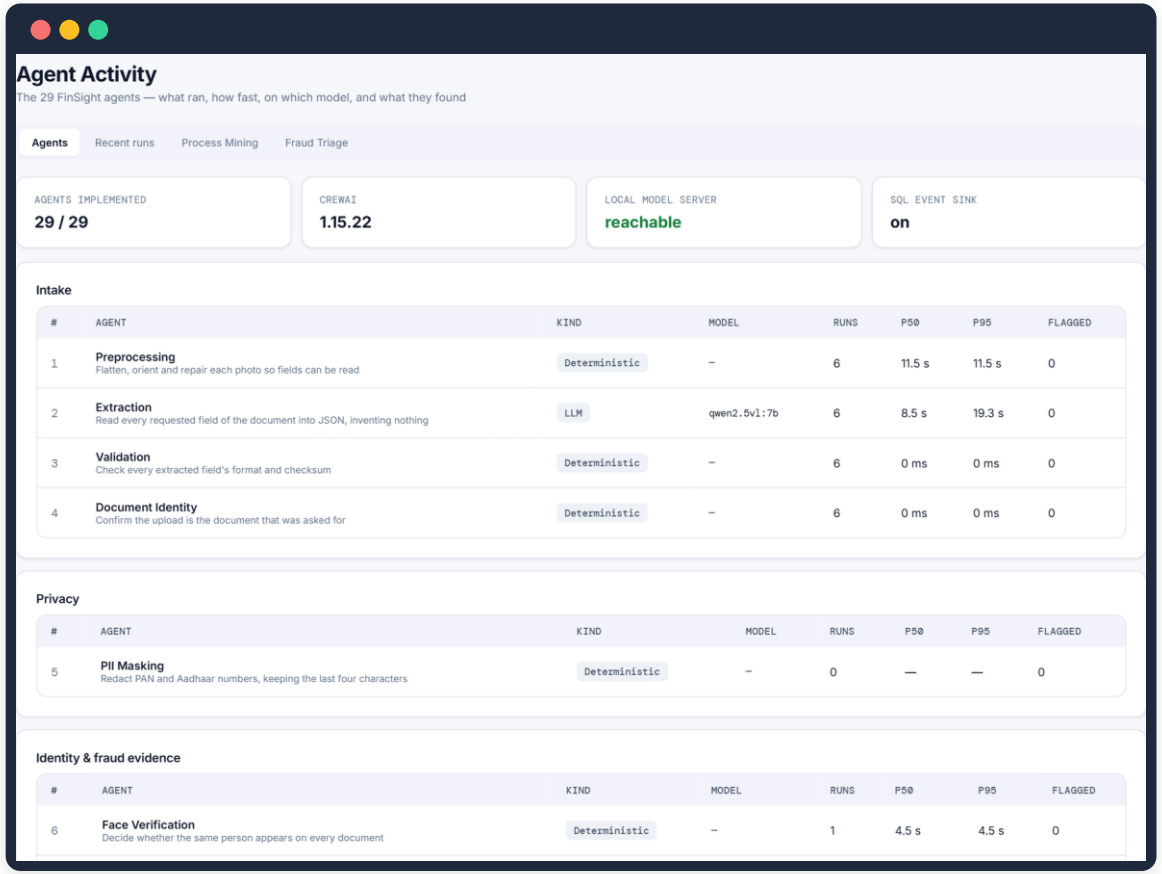
TRUST & CONTROL

Every step is traced — and becomes process-mining data

Runs and steps go to memory, JSON files and SQL Server. A database outage never fails a request.



Kill switch: AGENTS_ENABLED=false runs the same tools directly, untraced.



Agent Activity › Agents: each agent's kind, model, runs, p50 / p95 and flags.

TRUST & CONTROL

The database: SQL Server 'FinSight'

9 schemas · 82 tables · 17 views. Migrations live in docs/mssql/ — 08-agentic-layer.sql adds the agent tables.

core

9 tables

customers · loan_applications · uploaded_documents · document_extraction_results · loan_offers

fraud

9 tables

fraud_assessments · fraud_cases · fraud_rings · identity_keys · fraud_weight_sets

comp

5 tables

compliance_documents · document_chunks · rules · rule_candidates · check_results

risk

3 tables

risk_assessments · pricing_models · risk_thresholds

agent

4 tables

run · step · tool_call · llm_setting_audit

pm

20 tables

event_log · activity · case_metrics · case_variant · conformance_result · deviation

ops

8 tables

users · notifications · notification_templates · loan_products

audit

3 tables

audit_log · audit_log_details · business_events

ref

21 tables

application_status · lifecycle_transition · document_type · fraud_severity …

05

Evaluation

Every agent was tested as soon as it was built, against a pass mark written down before the run.



IN THIS SECTION

Method | Scorecard | Defects fixed | Latency | Limits & next steps

EVALUATION

How every agent was tested

Build → fix the target → run → record → write it up → next agent. A weak result is reported as it is.

Build the agent

Fix its target
`targets.yaml`

Run the eval
`run_eval --agent`

Save results
`results/<agent>.json`

Write the section
`report doc`

EVERY RUN RECORDS

- quality metric vs target
- p50 / p95 latency
- LLM calls made
- degraded / fallback rate
- failure examples, personal data removed

DATA USED

Synthetic generator
Ground-truth PAN, Aadhaar, payslips, statements, ITR, GST; forgeries; 54 fraud-pattern statements. The only data allowed to reach a hosted model.

16 real documents
Local model only; reported as aggregates. 9 genuine, 3 fabricated, 4 excluded.

45 public edited pairs
Original vs edited paystubs and statements — tampering on unfamiliar documents.

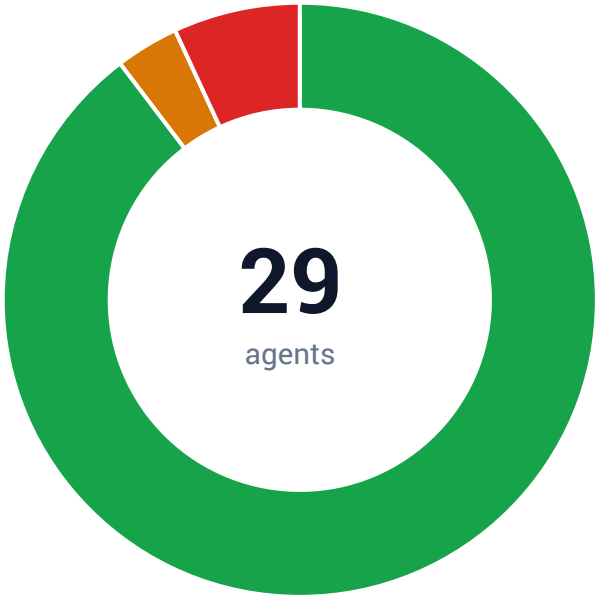
LFW public faces
Face verification, triage identities, liveness captures and simulated attacks.

50,000-loan NBFC book
Risk scoring on the held-out 20% only (anonymised).

RBI regulations + gold sets
40- and 12-question sets for knowledge base, rules, Q&A and policy.

EVALUATION

Scorecard: 26 of 29 agents meet their target



26	Strong
1	Acceptable
2	Weak

1 Preprocessing 99.6% target ≥ 85%	2 Extraction 97.2% target ≥ 90%	3 Validation 100% target 100%	4 Document Identity 100% target ≥ 90%	5 PII Masking 100% target 100%
6 Face Verification 99.4% target ≥ 90%	7 Tampering Forensics 100% target ≥ 85%	8 Authenticity 89.5% target ≥ 70%	9 Tamper Explainer 90% target ≥ 95%	10 Tamper Locator 33.3% target ≥ 80%
11 Cross-Doc Verification 100% target 100%	12 Dedupe 100% target ≥ 95%	13 Fraud Fusion 93% target ≥ 90%	14 Fraud Patterns F1 1.00 target ≥ 0.85	15 Fraud Triage 100% target ≥ 80%
16 KB Ingestion 91.7% target ≥ 85%	17 Rule Authoring 88.9% target ≥ 70%	18 Regulatory Q&A 100% target ≥ 85%	19 Policy Check 100% target 100%	20 Income Analysis F1 1.00 target ≥ 0.90
21 Risk Scoring AUC 0.771 target ≥ 0.75	22 Credit Memo 100% target ≥ 95%	23 UW Supervisor 100% target ≥ 85%	24 Offer Generation 100% target 100%	25 Workflow Router 100% target 100%
26 Notifier 100% target 100%	27 Audit / Storage 100% target 100%	28 Process Mining 100% target 100%	29 Liveness ACER 22.2% target ≤ 10%	

EVALUATION

Testing found — and fixed — 17 real defects

The eight that mattered most. Each was caught by an agent's evaluation or the latency benchmark, not by users.

PDF handling

Two threads inside PDFium at once crashed the whole server — with no error logged.

Fix One process-wide PDFium lock.

PII Masking

Sideways cards split the wrong way; one real PAN photo leaked in full.

Fix Upright rotation first: 0 leaks.

Document Identity

A payslip showing a PAN passed as a PAN — 54% of wrong uploads caught.

Fix Also reads the page's own words: 100%.

LLM gateway

'localhost' tried IPv6 first on Windows: +2 s on every model call.

Fix 127.0.0.1 and one pooled connection.

Risk Scoring

ML features built unlike training: held-out AUC 0.669.

Fix Training feature builder reused: 0.771.

Policy Check

Face rule used another model's 0.62 scale: every genuine match failed.

Fix SFace threshold 0.363.

Structured LLM calls

The model echoed the schema: 92% of written Q&A answers discarded.

Fix Schema-enforced JSON output.

Cross-Doc Verification

A shared surname alone counted as a name match — family members looked like one person.

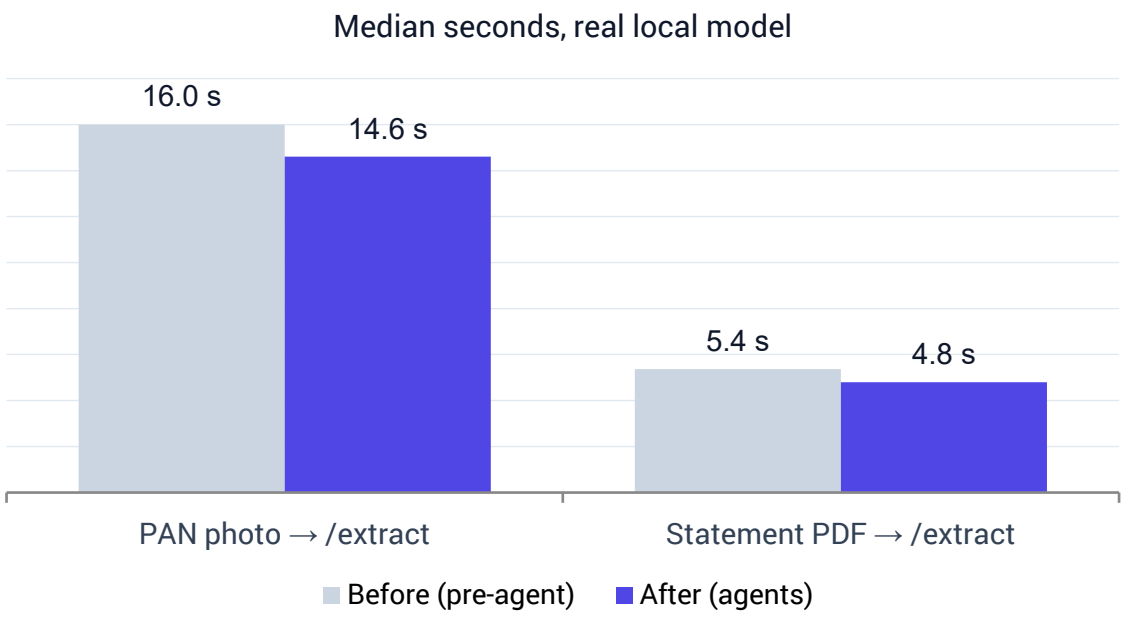
Fix Fixed in the browser and Python logic.

+ 9 more: card rules applied to full pages · images overflowing the model context · regulation titles ignored by grounding · unreadable photo rows treated as real · SQL inserts refused on tables with triggers · a parser error result · review statuses in the router · a stale Agent Activity tab · PDFs rasterised needlessly

EVALUATION

Faster than before the agents

The two uploads that use the local model got 9–11% faster; everything else added at most 56 ms.



Endpoint (model mocked)	Before	After	Change
`/extract · PAN photo`	4.33 s	4.13 s	**−202 ms**
`/extract · statement PDF`	11 ms	33 ms	**+22 ms**
`/document/integrity · PDF`	62 ms	70 ms	**+8 ms**
`/document/integrity · photo`	1.14 s	1.19 s	**+56 ms**
`/document/mask · PAN`	1.29 s	0.85 s	**−441 ms**
`/face/verify · two cards`	631 ms	641 ms	**+10 ms**

2 s per model call

Windows tried IPv6 for 'localhost' first — now 127.0.0.1, pooled.

Photos 19% slower

A second OCR pass on ID cards — now reuses preprocessing's words.

Statement photos ~23 s

Their rows are now parsed in the background after the upload.

EVALUATION

Known limits and what comes next

LIMIT	NEXT STEP
Tampering on genuine photos 17% of genuine camera-style photos score HIGH	Recalibrate ELA on real phone photos
Tamper Locator · Weak 33% Boxes only edited figures, not copy-move or photo swaps	Turn keypoint clusters and photo regions into boxes
Liveness · Weak ACER 22% Only public photos and simulated attacks to test on	Collect real selfies and real print / screen attacks
Tamper Explainer · 90% 4 of 40 explanations cited a figure not in the findings	Tighten the prompt (the gate already hides them)
Real-world data No real fraud rings; income and patterns tested on synthetic data	Label a real production sample
Timings The 8 GB GPU was shared with other programs	Re-measure on a dedicated GPU

NEEDS YOUR INPUT

1. **Confirm Fraud Fusion v2**
   93% vs 60% for the original design; switching back is one line.
2. **Add an Anthropic key**
   Runs the Local-vs-Claude comparison on synthetic data only.
3. **Share real selfies**
   A small in-house set to calibrate the liveness threshold.

GETTING STARTED

Run it yourself

Five steps on one Windows machine. Ports: frontend 5173, API 8000, Ollama 11434.

1 Database

```
run docs/mssql/*.sql
on (localdb)\MSSQLLocalDB
```

2 Local model

```
ollama pull qwen2.5vl:7b
```

3 API :8000

```
cd api
python -m venv .venv-agents
.venv-agents\Scripts\pip install -r requirements-agents.txt
run.bat
```

4 Frontend :5173

```
npm install
npm run dev
```

5 Tests & evaluation

```
pytest tests/unit tests/agents tests/contract
python -m eval.run_eval --all
```

WHERE THINGS LIVE

api/app/agents/ registry, agent implementations, flows, tools, llm/gateway.py

api/eval/ datasets, targets.yaml, run_eval.py, results/

docs/finsight-architecture.html 10 diagrams incl. the agent layer

docs/reports/06-…Evaluation-Report.docx the full per-agent report

FinSight AI

Agents read, check and draft. People decide.

29	26	17	0
agents in 6 crews	meet their target	defects found & fixed	auto-rejections

Local model by default · every number grounded · every step traced · every decision human

Thank you · Questions?